

2. Parameters

Every arbitrary constant in this procedure is a named parameter listed here. The suggested values are starting points for the VSX3000 at 500–1000 mm. Values marked † depend on VSX3000 datasheet or SDK values that are not available at the time of writing, so they must be confirmed or replaced in Step 4.5.

Name	Suggested value	Units	Used in	Meaning
Sensor geometry				
SENSOR_FX_PX, SENSOR_FY_PX	from SDK †	px	all	Left IR focal lengths
SENSOR_CX_PX, SENSOR_CY_PX	from SDK †	px	all	Left IR principal point
SENSOR_BASELINE_MM	from datasheet †	mm	C, E	Stereo baseline B
PROJECTOR_OFFSET_MM	from datasheet †	mm (H, V, Z)	C, E	Projector center relative to the left camera
DEPTH_LSB_MM	from SDK †	mm	A, B-Z	Least significant bit of the depth output
Working volume				
Z_MIN_MM / Z_MAX_MM	500 / 1000	mm	all	Range limits
Z_NOISE_STEP_MM	50	mm	A	Spacing of noise stations, giving 11 stations
Z_SHAPE_STATIONS_MM	500, 625, 750, 875, 1000	mm	B, C, D	Stations for edge, area, and detection tests
Z_REDUCED_STATIONS_MM	500, 750, 1000	mm	B-Z, D (0% series)	Subset used for the slowest tests
Z_REFERENCE_MM	750	mm	4, 5, 7, 8	Mid-range station used for the warm-up check, the drift sentinels, the re-mount check, the C field sub-series, and the D pilot
FIELD_OFFSET_FRACTION	0.6		A, C	

		of half-FOV		Off-axis field positions at ($\pm f \cdot W/2$, $\pm f \cdot H/2$) from center
TILT_ANGLES_DEG	0, 15, 30, 45	deg	A	Plate tilts about H and about V; the 0° pose repeats the main-series center pose as a deliberate repeat visit
Capture control				
WARMUP_MIN_MINUTES	45	min	4	Minimum powered time before any capture
WARMUP_DRIFT_WINDOW_MIN	10	min	4	Window over which drift is judged
WARMUP_CHECK_INTERVAL_MIN	1	min	4	Interval between warm-up check captures
WARMUP_CHECK_FRAMES	10	frames	4	Frames per warm-up check capture
WARMUP_DRIFT_FRACTION_OF_SIGMA	0.1	—	4	Allowed drift of the mean plane Z, as a fraction of σ_t at that Z
DRIFT_SENTINEL_INTERVAL_MIN	60	min	all	Interval between sentinel captures
SENTINEL_FRAMES	30	frames	all	Frames per drift sentinel capture
ROBOT_SETTLE_TIME_S	2.0	s	all	Wait after motion before capture; Step 4.4 verifies it
SETTLE_CHECK_FRAMES	100	frames	4	Frames per capture of the settle and vibration check
SETTLE_SIGMA_EXCESS_FRACTION	0.10	—	4	Servo-on σ_t may exceed brakes-on σ_t by at most this fraction
FRAMES_PER_NOISE_STATION	100	frames	A	Frames per Z and field pose of the main A series
FRAMES_PER_TILT_POSE	50	frames	A	Frames per pose of the tilt sub-series
FRAMES_PER_EDGE_POSE	30	frames	B-HV	Frames per edge pose
FRAMES_PER_ZSTEP_POSE	10	frames	B-Z	Frames per Z-step pose
FRAMES_PER_AREA_POSE	10	frames	C	Frames per area pose
FRAMES_PER_DETECTION_TRIAL	1	frames	D	One frame per trial, giving single-frame detectability
POSE_LOG_RESOLUTION_MM	0.01	mm	all	Minimum resolution of the logged read-back robot pose
FRAME_CHECK_PX	0.5	px	4	Required agreement between the IR-image edge and the depth discontinuity in Step 4.5
FRAME_RATE_HZ	from SDK † (10 assumed)	frames/s	9	Frame rate in the chosen trigger mode; the capture budget uses it

MOVE_AND_SETTLE_TIME_S	3	s	9	Robot move plus settle time per pose assumed by the capture budget
Phase randomization				
PHASE_JITTER_SPAN_PX	8 †	px at the station Z	B, C, D	Square span of random lateral offsets. Must exceed the projector dot pitch and the noise correlation length, neither of which is known before A is analyzed; Section 10 Step 7 checks it
PHASE_JITTER_POSES_EDGE	25	poses	B-HV	Offsets per edge station
PHASE_JITTER_POSES_AREA	30	poses	C	Offsets per area station
FIELD_SUBSERIES_POSES_AREA	10	poses	C	Random-offset poses per array at each off-axis field position
Z-step test				
Z_STEP_LADDER_MM	0.1, 0.2, 0.5, 1, 2, 4	mm	B-Z	Commanded step sizes
Z_STEP_REPEATS	10	ABAB cycles	B-Z	Alternations for each step size
Z_STAIRCASE_SUBDIVISION	10	per quantum	B-Z	Fine-sweep points per expected quantum
Z_STAIRCASE_QUANTA	3	quanta	B-Z	Extent of the fine staircase above Z_0 , in expected quanta δZ_q
Z_STAIRCASE_FRAMES	10	frames	B-Z	Frames per fine-staircase step
ZSTEP_PATCH_SIZES_PX	1, 5, 20	px (square side)	B-Z	Patch sizes over which the A → B difference is averaged
ROBOT_REPEATABILITY_MM	0.05	mm	B-Z	Position repeatability of the robot (ISO 9283, the Section 3.1 requirement), taken as the error of the read-back pose that serves as the B-Z step truth. The ratio of this value to each rung (50% at the 0.1 mm rung) is reported with δ_{50} as the truth uncertainty
Z_STEP_APPROACH_OVERSHOOT_MM	2	mm	B-Z	Every Z visit is approached from below by this overshoot and a final move up onto the pose, so that backlash and compliance do not enter the A → B difference
Detection (D, also used in B-Z)				
DETECTION_FALSE_ALARM_TARGET	0.01	per window	B-Z, D	

				Sets threshold τ from the null distribution: blank sites in D, A $\rightarrow$ A visit differences in B-Z
DETECTION_MIN_CONNECTED_PX	2	px	D	Minimum connected region counted as a detection
DETECTION_WINDOW_MARGIN_PX	3	px	D	Search window radius equals D/2 in px plus this margin
DETECTION_LEVELS	9	diameters	D	Diameter levels per curve
DETECTION_LEVEL_LOW_FACTOR / HIGH_FACTOR	0.3 / 2.5	$\times$ pilot D_50	D	Range spanned by the levels, wide enough to include the 0% and 10% points
DETECTION_FINE_LADDER_RATIO	$2^{(1/4)}$	—	3, D	Diameter ratio of a dedicated fine detection plate, used when the $\sqrt{2}$ ladder gives too few levels
DETECTION_TRIALS_PER_LEVEL	60	trials	D	Independent trials per level
DETECTION_ZERO_TRIALS	300	trials	D	Trials at the levels next to the 0% point
DETECTION_ZERO_PROBABILITY_BOUND	0.01	—	D	Upper bound that defines practically zero detection
DETECTION_LAPSE_RATE_MAX	0.05	—	B-Z, D	Upper bound on the lapse rate λ of the psychometric fit
INDEPENDENCE_SIGMA_MULTIPLE	2	$\times 1/\sqrt{n}$	D	Lag-1 autocorrelation of a detection sequence must lie within this many standard errors of zero
CONTINUOUS_ANGLE_STEP_MM	10	mm	D	Z step of the optional continuous-angle variant
CONTINUOUS_ANGLE_POSES_PER_STEP	20	poses	D	Random-offset poses per Z step of that variant
CONFIDENCE_LEVEL	0.95	—	B, C, D, E	For all confidence intervals and bounds
BOOTSTRAP_RESAMPLES	2000	—	C, D, E	Resamples for confidence intervals
Noise, edge, and area analysis (A, B, C)				
NOISE_CLOSURE_TOLERANCE	0.2	—	A	Allowed relative mismatch of σ_{tot}^2 against $\sigma_{\text{t}}^2 + \sigma_{\text{fp}}^2 + \text{bias}^2$
QUANTIZATION_PATCH_PX	20	px (square side)	A	Central patch whose raw depth codes are histogrammed for the quantum
AUTOCORRELATION_THRESHOLD	1/e	—	A	Normalized autocorrelation level that defines the correlation length

LEGACY_BOX_HALF_PX	10	px	A	Half side of the 20 × 20 px boxes of the existing testZRepeatabilityBrownBoard.py metrics
LEGACY_METRIC_DEPTHS_MM	700, 1000	mm	A	Depths at which the legacy metrics are computed
ESF_RISE_LOW / ESF_RISE_HIGH	0.1 / 0.9	of step height	B-HV	Normalized heights that bound the rise distance
ESF_HALF_HEIGHT	0.5	of step height	B-HV, C	Normalized height of the edge crossing s_50 and of the area iso-contour
ESF_AGREEMENT_BINS	1	bins	B-HV	Required agreement between the robot-stepped and slanted-edge ESFs once aligned
ESF_LINEARITY_TOLERANCE_H	0.1	of step height	B-HV	Largest allowed difference between the small-gap and large-gap ESFs before resolution is called step-height dependent
FRONT_READ_HEIGHT_THRESHOLD	0.5	of step height	C	Normalized height above which a read counts as a front read
Boundary bias (E)				
BOUNDARY_BAND_HALF_WIDTH_PX	8	px	E	Analysis band on each side of a true edge
BOUNDARY_BIN_WIDTH_PX	0.25	px	B, E	Signed-distance bin width, giving 4× oversampling
SURFACE_ASSIGNMENT_SIGMA_MULTIPLE	3	× $\sigma_{\text{tot}}(Z)$	E	A read within this distance of a reference plane is assigned to that plane
Targets				
NOISE_PLATE_SIZE_MM	400 × 400	mm	A, B-Z	Uniform matte plate
PLATE_FLATNESS_MM	0.05 †	mm	all	Must be at most PLATE_FLATNESS_SIGMA_FRACTION × the smallest expected σ_{tot} ; the value assumes $\sigma_{\text{tot}} \geq 0.2$ mm at Z_MIN
PLATE_FLATNESS_SIGMA_FRACTION	0.25	—	3	Flatness limit as a fraction of the smallest expected σ_{tot}
GAP_SMALL_MM / GAP_LARGE_MM	15 / 60	mm	B, C, D, E	Distance from front plate to back plate
EDGE_SLANT_DEG	5	deg	B-HV	Edge rotation from the image axes
EDGE_SQUARE_SIZE_MM	160	mm	B-HV	Side of the raised square and of the square window

CHAMFER_MARGIN_DEG	10	deg	3	Margin added to the worst-case ray angle
EDGE_LAND_MAX_MM	0.2	mm	3	Maximum residual flat land at a knife edge
DIAMETER_LADDER_RATIO	$\sqrt{2}$	—	C, D	Ratio between successive diameters
DIAMETER_MIN_FOOTPRINT_FRACTION	0.3	$\times$ $p(Z_{\text{MIN}})$	C, D	Smallest diameter, as a fraction of the pixel footprint at Z_{MIN}
DIAMETER_MAX_FOOTPRINT_MULTIPLE	30	$\times$ $p(Z_{\text{MAX}})$	C	Largest diameter, as a multiple of the pixel footprint at Z_{MAX}
FEATURE_ISOLATION_PX	30	px at Z_{MIN}	C, D	Minimum edge-to-edge spacing between features
POST_DIAMETER_FRACTION_OF_D0	0.5	$\times$ pilot D_0	3	Disk support posts must be thinner than this. Before fabrication D_0 is estimated from the Tier-A model; the D pilot and the post-only control sites confirm the choice
Registration				
REGISTRATION_POSES	30	poses	4	Hand-eye poses spanning the volume
REGISTRATION_TILT_RANGE_DEG	± 20	deg	4	Tilt range about H and V
REGISTRATION_FRAMES	10	frames	4	Frames per registration pose
REGISTRATION_RESIDUAL_ACCEPT_MM	0.15	mm RMS	4	Acceptance limit for the registration residual

These relations are used throughout. $p(Z)$ is the pixel footprint, θ the subtended angle, d the disparity, and δZ_q the depth quantum implied by a disparity quantum q :

$$p(Z) = \frac{Z}{f_x}, \quad \theta = \frac{D}{Z}, \quad D_{\text{px}} = \frac{D}{f_x Z}, \\ \quad d = \frac{k}{Z}, \quad k = f_x B, \quad \delta Z_q = \frac{q}{Z^2} k$$